ED ML LoS vs Competitor Gate Engine — One-Pager

Purpose
Give decision-makers a crisp comparison between the **ML-only LoS→occupancy** approach and the **MLP + deterministic gate engine** approach, plus a recommended hybrid.

Snapshot

Your project — ML LoS → Occupancy - Predict **patient-level LoS** (CatBoost) from ED features and **aggregate** to hospital-wide bed forecasts (24–72h).
- Key insight: **error symmetry** in LoS often improves occupancy more than simply lowering MAE.
- Future arrivals filled with **averages**; evaluates hospital-wide occupancy (not ward-specific).

Competitor — MLP + Gate Engine - ML outputs **admission risk + LoS buckets**; a **deterministic gate engine** turns this into **policy-bound actions** (priority, next_action, TTL, reason).
- ML is **advice-only**; gates enforce **safety & explainability**; includes ops panel/nudges, label spec for an actionable target.

Head-to-Head

Dimension	ML LoS → Occupancy	MLP + Gate Engine
Primary output	Beds forecast (H=24/48/72h)	Actionable queue rules (priorities, timers, reasons)
Unit of decision	Hospital level (aggregated)	Patient/task level (operational)
Future arrivals	Constant average fill (calendar effects missing)	Not modeled yet (focus on gates)
Granularity	Hospital-wide; no ward/acuity isolation	Per-patient policy; no occupancy total
Governance	No rule layer; no fairness audit	Read-only ML; deterministic gates + guardrails
Calibration	Not reported; post-hoc symmetrization sims	Calibrator in place; shift-aware fallbacks
Data hygiene	Standard ED features; long-stay truncation	Explicit no-leakage (features $\leq t_0$)

Dimension	ML LoS → Occupancy	MLP + Gate Engine
Ops readiness	Good for capacity planning; lacks ops hooks	Good for day-to-day flow; lacks occupancy

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Strengths & Gaps

ML LoS → Occupancy
+ Converts patient predictions into bed needs; quantifies what error properties matter.
− Average-fill arrivals, no ward/acuity split, tail truncation; no rule/governance layer.

MLP + Gate Engine
+ Clear safety contract; policy knobs; explainability; ops panel/nudges.
− No hospital-wide occupancy; some label/calibration maturity still needed.

When to Use Which

- **Bed planning 24–72h out:** choose **ML LoS→occupancy** (add calendar-aware arrivals).
- **Running the ED today (escalations, timers, nudges):** choose **Gate Engine** (ML as calibrated hints).

Recommended Hybrid (Best of Both)

1) **Keep the occupancy aggregator**; replace average-fill with **calendar-aware arrivals** (day-of-week/season/case-mix).
2) **Adopt the gate engine** as a policy layer: define a clear positive label/time window (Gate_Pos v2), costs, exclusions; keep ML **read-only**.
3) **Calibrate & audit**: reliability curves, Brier/ACE, and **fairness slices** (sex, age, acuity, arrival hour, language, insurance).
4) **Wire together**: feed **LoS quantiles + admit risk** into gates; publish **ward/acuity/isolation** occupancy for staffing.

Pilot & Metrics

Shadow-mode → capped pilot with governance sign-off. Track: - **Beds MAE@H** (H=24/48/72) per ward + P95, **overflow/shortfall days**, signed bias.
- **Nudge utility:** TP − λ·FP − μ·FN; clinician "Not helpful" rate; override counts.
- **Flow:** boarding time, diversion hours, elective cancellations avoided.
- **Fairness:** error/calibration parity across subgroups.

Decision Guidance (TL;DR)

- Need **explainable, policy-controlled actions now** → start with **Gate Engine**.
- Need **reliable capacity forecasts** → lean on **ML LoS→occupancy** with better arrivals.
- Need **both** → implement the **hybrid** above; pilot in shadow-mode, then A/B with decision-centric metrics.